

Project Proposal: Visualizing Water in the Colorado River Basin

Introduction

The Colorado River is a critical water source for over 40 million people across seven U.S. states and Mexico, supporting agriculture, hydropower, and urban water supplies. However, prolonged drought, increasing demand, and climate change have significantly reduced water availability. Understanding spatial and temporal trends in water scarcity is essential for effective resource management and policy decisions. This project aims to create an interactive visualization of water trends in the Colorado River Basin, integrating historical flow data, reservoir levels, and climate variables (U.S. Bureau of Reclamation, 2021).

Problem Statement

Water levels in the Colorado River and its reservoirs, such as Lake Powell and Lake Mead, have reached historic lows, threatening water security for millions (NASA Earth Observatory, 2022). Current decision-making often relies on static reports and datasets, limiting accessibility for policymakers, researchers, and the public. A dynamic, GIS-based visualization of changing water availability will help communicate the severity of the crisis and support informed decision-making.

Proposed Methodology

The project will begin with data collection from various sources. Streamflow data for major river segments and reservoir levels will be gathered from the U.S. Bureau of Reclamation and the U.S. Geological Survey. Additionally, water use data from state water agencies will be incorporated to assess demand and consumption patterns.

Following data collection, the processing and analysis phase will involve aggregating and cleaning historical data on river flow and reservoir levels. Climate trends will be correlated with water availability to identify long-term patterns. Spatial and statistical analysis will be conducted using RStudio and GIS software, specifically ArcGIS Pro.

The final phase of the project will focus on visualization and dashboard development. An interactive ESRI Story Map will be created to illustrate water scarcity trends in the Colorado River Basin. Temporal animations will be integrated to depict changes in reservoir levels and precipitation anomalies over time. Additionally, interactive graphs allow users to explore the dynamic relationship between water use and availability. This visualization will serve as an informative tool for policymakers, researchers, and the public to understand better and address the region's water scarcity challenges.

Expected Results

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This project will produce a dynamic GIS-based dashboard visualizing changes in water availability in the Colorado River Basin. The visualization will illustrate trends in declining reservoir levels, variations in precipitation and snowpack, and increasing water demand. The final product will be an educational tool for policymakers, researchers, and the public, fostering a clearer understanding of the challenges facing the region's water resources.

References

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